Ali AGHABABAEI

Room 324, Laboratoire d'Informatique de Grenoble, University of Grenoble Alpes Email

+33 638 233343 aah94.github.io LinkedIn Scholar

Summary

I am a final-year Ph.D. candidate at Université Grenoble Alpes, supervised by Professor Massih-Reza Amini. My research explores model efficiency and representation learning within machine learning and deep learning, with the goal of designing scalable and practical AI systems.

Education

Université Grenoble Alpes *2023 - Present*
Ph.D. in Computer Science

- **Research:** Deep Learning Efficiency, Representation Learning

Sharif University of Technology *2018 – 2022*
M.Sc. in Computer Science

- **Coursework:** Deep Learning, Speech Processing

University of Tehran *2013 – 2018*
B.Sc. in Electrical Engineering

- **Coursework:** Linear Algebra, Pattern Recognition, Digital Signal Processing

Research Interests

Deep Learning Efficiency

- Tensor Decomposition
- Knowledge Distillation
- Memory-Efficient Training

Representation Learning

- Semi-Supervised Learning
- Pseudo-Labeling
- Contrastive Learning

Generative AI

- Probabilistic Modeling
- Image Denoising
- Image Super-Resolution

Research Experience

Ph.D. Thesis *2023 - Present*
APTIKAL team

- Developing a new method for **memory-efficient** training by improving greedy supervised learning. Proposed a new method for preserving information during local training using neural information estimation.
- Extend the **Forward-Forward** algorithm by scaling it to Vision Transformers, introducing early cross-attention between label tokens and feature representations.
- Investigated semi-supervised learning, focusing on **pseudo-labeling** through a theoretical analysis of bias stability. Proposed an adaptive **Gaussian geometric** loss to shape latent representations, accelerate convergence, and reduce majority-class bias.
- Reduced the computational complexity and parameter count of pre-trained deep learning models by developing a **tensor decomposition** approach, utilizing a custom **constraint-based** loss function to identify optimal ranks while preserving accuracy.

Research Collaborator *2021 – 2022*
University of Basel

- Conducted analysis of inverse problems, including image denoising, limited-view computed tomography (CT), and wave scattering, utilizing invertible neural networks.
- Approximated data distribution with normalizing flows to identify out-of-distribution samples, improving model robustness and reliability.

M.Sc. Thesis *2018 – 2021*
Electronic Research Institute

- Proposed a patch-wise feature analysis approach for identifying forgery in video frames, aimed at enhancing deepfake detection capabilities.
- Developed a straightforward method to enhance the generalization and robustness of deepfake detection models.

B.Sc. Thesis *2016 – 2017*
University of Tehran

- Designed and implemented a video quality meter to assess video quality in terms of blockiness and blurriness distortions in a no-reference mode.

Publications

JEPAMatch: Geometric Representation Shaping for Semi-Supervised Learning *Mar 2026*
Ali Aghababaei-Harandi, Aude Sportisse, Massih-Reza Amini
arXiv preprint

Unified Framework for Pre-trained Neural Network Compression via Decomposition and Optimized Rank Selection *Sep 2025*
Ali Aghababaei-Harandi, Massih-Reza Amini
ECML-PKDD 2025

Deep Variational Inverse Scattering *Mar 2023*
AmirEhsan Khorashadizadeh, ***Ali Aghababaei-Harandi***, Tin Vlašić, Hieu Nguyen, Ivan Dokmanić
EuCAP 2023

Conditional Injective Flows for Bayesian Imaging *Feb 2023*
AmirEhsan Khorashadizadeh, Konik Kothari, Leonardo Salsi, ***Ali Aghababaei-Harandi***, Maarten de Hoop, Ivan Dokmanić
IEEE Transactions on Computational Imaging

Talks & Presentations

Efficient Deep Learning *2025*
Mathematical Fundamentals of AI Workshop, Sorbonne Center for AI (SCAI), Paris, France

Presentation on Unified Framework for Pre-trained Neural Network Compression *2025*
European Conference on Machine Learning (ECML-PKDD), Porto, Portugal
Conférence sur l'Apprentissage Automatique (CAp), Dijon, France

Skills

Programming: Python, PyTorch, TensorFlow, JAX, Hugging Face, Git

Theoretical: Pattern Recognition, Design and Analysis of Algorithms, Deep Learning